Statistics Review

CUHK, Shenzhen

ECO3121

Causality

- We are interested in causal relations, but statistics only allows us to establish correlations
- To say that one variable has a causal effect on another, other factors affecting the outcome must be held fixed
- Sometimes can use controlled experiments
- Experiment are often impossible in economics (too costly and/or for ethical reasons)
- Must rely on observational data

- Example: effect of health insurance coverage on health
- Question: what is the effect of health insurance on health?
- Ideal experiment: randomly give people health insurance or not, compare their health afterward
- Observed survey data:

Group	Sample Size	Mean Health	Std.Dev.
Some insurance	8114	4.01	0.93
No insurance	1281	3.70	1.01

Source: 2009 NHIS data reported by **?**

- Is 4.01 - 3.70 = 0.31 the causal effect of health insurance on health?
- Not likely, people with and without insurance differ in other ways that might affect health

Group	Mean Education	Mean income
Some insurance	14.31	106,467
No insurance	11.56	45,656

Potential Outcomes

- Potential outcomes (aka the Rubin causal model) are one powerful way of thinking about causality
- Let $D_i = 1$ if person i has health insurance, 0 otherwise
- Let Y_{i0} and Y_{i1} be the **potential outcomes**
  - Y_{i0} = health if person i does not have health insurance
  - Y_{i1} = health if person i has health insurance
- We observe

$$
\begin{aligned}
Y_i &= \begin{cases} Y_{i0} & \text{if } D_i = 0 \\ Y_{i1} & \text{if } D_i = 1 \end{cases} \\
&= (1 - D_i) Y_{i0} + D_i Y_{i1} \\
&= Y_{i0} + D_i \times \underbrace{(Y_{i1} - Y_{i0})}_{\text{causal effect of health insurance for person } i}
\end{aligned}
$$

Potential Outcomes (continued)

- Observed average difference in health

$$\begin{aligned}\mathrm{E}[Y_{i1}|D_i=1]-\mathrm{E}[Y_{i0}|D_i=0] &= \\ &= \underbrace{\mathrm{E}[Y_{i1}-Y_{i0}|D_i=1]}_{\text{Average treatment effect on the treated}} + \\ &\quad + \underbrace{\mathrm{E}[Y_{i0}|D_i=1]-\mathrm{E}[Y_{i0}|D_i=0]}_{\text{Selection bias}}\end{aligned}$$

- Insurance may help you or hurt you (on average), but wealthier people are more likely to afford insurance and might be healthier for other reasons; conversely people who are sick might be more willing to pay for insurance
- Random assignment of D_i removes selection bias because D_i and Y_{i0} are then independent
- Often cannot randomize, but can
  - Use available data to approximate desired experiment
  - Use economic theory to impose restrictions

Probability

- Purpose: system for quantifying chance and making predictions about future events
- Interpretations:
  - Relative frequency of an event in many repeated trials
  - Subjective assessment of the likelihood of an event
- The properties of probability are fairly intuitive, and you can work with probabilities without worrying too much about how probability is formally defined. However, if we want to be mathematically rigorous, we should carefully define what we mean by probability.

Basic definitions

- Random experiment: procedure that has well-defined set of outcomes and "could" be infinitely repeated
- Sample space: set of possible outcomes of an experiment, $S = \{a_1, a_2, ..., a_J\}$
- Event: any subset of sample space, $A \subseteq S$
- Probability: function from all subsets of the sample space, S, to $[0, 1]$ such that
  1. $\mathrm{P}(S) = 1$
  2. $1 \geq \mathrm{P}(A) \geq 0$ for all $A \subseteq S$
  3. If $A_1, A_2, ...$ are disjoint events then $\mathrm{P}(A_1 \cup A_2 \cup ...) = \mathrm{P}(A_1) + \mathrm{P}(A_2) + ...$
- Random variable: function from S to a number

Random variable

An example of a random experiment is rolling a dice. In this case the sample space is $S = \{1, 2, 3, 4, 5, 6\}$. Some events are: get a $1 = \{1\}$, get more than 4, $\{5, 6\}$, etc.
A random variable is a numeric representation of a random experiment. For example, our experiment could be flipping a coin. Then $S = \{\text{heads}, \text{tails}\}$.
A function assigns a number to each element of S, so one example of a random variable is $X = \begin{cases} 1 & \text{if heads} \\ 0 & \text{if tails} \end{cases}$.

Another is $Y = \begin{cases} 50 & \text{if heads} \\ -257 & \text{if tails} \end{cases}$.

A third example is $Z = \begin{cases} 1 & \text{if heads} \\ 1 & \text{if tails} \end{cases}$.

Properties of probability

From the definition of probability on the previous slide, we can derive some intuitive properties.

1. $\mathrm{P}(\emptyset) = 0$
2. $\mathrm{P}(A^c) = 1 - \mathrm{P}(A)$
3. $A \subseteq B$ implies $\mathrm{P}(A) \leq \mathrm{P}(B)$
4. $\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$

Conditional probability

Conditional probability is very important for making decisions and interpreting our observations of the world. Often, we would like to know about something that we do not directly observe, but we do observe some related information. For example, to forecast GDP next quarter, which is something we can not observe today, we can use current information about GDP, inflation, consumer confidence, etc. To make a forecast, we do not care so much about the unconditional probability that GDP tomorrow is high, but about the probability that GDP tomorrow is high given what we observe today.

Conditional probability

- Conditional probability: $\mathrm{P}(A|B) = \mathrm{P}(A \cap B)/\mathrm{P}(B)$
  - Satisfies axioms of unconditional probability (and so also has properties on previous slide)
- Bayes' Rule $\mathrm{P}(A|B) = \frac{\mathrm{P}(B|A)\mathrm{P}(A)}{\mathrm{P}(B)}$
- A is independent of B iff $\mathrm{P}(A \cap B) = \mathrm{P}(A)\mathrm{P}(B)$, denote as $A \perp\!\!\!\perp B$
  - $A \perp\!\!\!\perp B$ implies $\mathrm{P}(A|B) = \mathrm{P}(A)$ and $\mathrm{P}(B|A) = \mathrm{P}(B)$.

Bayes' Rule

Proof of Bayes' rule:
Definition of conditional probability: $\mathrm{P}(A \cap B) = \mathrm{P}(A|B)\mathrm{P}(B)$,
and switching role of A and B, $= \mathrm{P}(B|A)\mathrm{P}(A)$.
Rearrange to get conclusion.

Bayes' Rule

Bayes' Rule allows us to switch between the $\mathrm{P}(A|B)$ and $\mathrm{P}(B|A)$. It is very useful. One common use is for interpreting the results of screening tests. For example, suppose there is a test for some disease. From clinical trials, we would likely know the probability of correctly testing positive conditional on having the disease, $\mathrm{P}(+|\mathrm{disease})$, and the probability of falsely testing positive, $\mathrm{P}(+|\mathrm{healthy})$. Suppose the unconditional probability of having the disease if $\mathrm{P}(\mathrm{disease})$.

Bayes' Rule

From this, we can use Bayes' rule to calculate the probability of having the disease conditional on testing positive as:

$$\begin{aligned}\mathrm{P}(\text{disease}|+) &= \frac{\mathrm{P}(+|\text{disease})\mathrm{P}(\text{disease})}{\mathrm{P}(+)} \\ &= \frac{\mathrm{P}(+|\text{disease})\mathrm{P}(\text{disease})}{\mathrm{P}(+|\text{disease})\mathrm{P}(\text{disease}) + \mathrm{P}(+|\text{healthy})(1-\mathrm{P}(\text{disease}))}.\end{aligned}$$

Remark: When A and B are independent, then knowing whether B happens does not give you any information about the probability of A occurring

Discrete random variables

- Recall: a random variable is a function from S to a number
- A random variable is discrete if it can only take countably many different values
- E.g. $X \in \{x_1, ..., x_m\}$
- Probability mass function (PMF) of X, $p_i = \mathrm{P}(X = x_i)$ gives the probability that X equals each of its possible values
- Cumulative distribution function (CDF): $F(x) = \mathrm{P}(X \leq x) = \sum_{i=1}^{m} p_i 1\{x_i \leq x\}$

Continuous random variables

- A random variable is continuous if it takes on any single real value with zero probability
- Set of possible values is uncountably infinite (e.g. real line or line segment)
- CDF is continuous and differentiable
- Probability density function (PDF): the derivative of the CDF

$$f(x) = \frac{dF}{dx}(x) \quad \text{and} \quad F(x) = \int_{-\infty}^{x} f(t)dt$$

Bivariate distributions

- Discrete:
  - Joint PMF $f_{X,Y}(x,y) = \mathrm{P}(X = x, Y = y)$
  - Marginal PMF
    $f_X(x) = \mathrm{P}(X = x) = \sum_{y \in \text{all values of } Y} \mathrm{P}(X = x, Y = y)$
  - Conditional PMF $f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$
- Continuous:
  - joint CDF
    $F_{X,Y}(x,y) = \mathrm{P}(X \leq x, Y \leq y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(u,t) du dt$
  - joint PDF $f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}}{\partial x \partial y}(x,y)$
  - Marginal CDF for X, $F_X(x) = \mathrm{P}(X \leq x) = F_{X,Y}(x,\infty)$
  - Marginal PDF for X, $f_X(x) = \frac{dF_X}{dx}(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy$
  - Conditional PDF $f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$

Independence

- Random variables X, Y are independent if $f_{X,Y}(x, y) = f_X(x) f_Y(y)$ for all x, y
- $X \perp\!\!\!\perp Y$ implies $f_{X|Y}(x|y) = f_X(x)$

Expectation

We will use expectations a lot, so it's important to understand them and know their properties. The fact that the expectation is linear and the law of iterated expectations will be especially useful.

Expectation

- The expectation of X is

$$\mathrm{E}[X] = \begin{cases} \sum_{i=1}^{m} x_i f_X(x_i) & \text{if } X \text{ discrete} \\ \int_{-\infty}^{\infty} x f_X(x) dx & \text{if } X \text{ continuous} \end{cases}$$

  - Is a constant
  - a.k.a. expected value, population average, population mean, first moment
- Expectation is linear:

$$\mathrm{E}[aX + bY] = a\mathrm{E}[X] + b\mathrm{E}[Y]$$

- Expectation of function g:

$$\mathrm{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

Variance and other moments

- kth moment of X: $\mathrm{E}[X^k]$
- kth central moment of X: $\mathrm{E}\left[(X - \mathrm{E}[X])^k\right]$
- Variance is the 2nd central moment

$$\begin{aligned}\mathrm{Var}(X) =& \mathrm{E}\left[(X - \mathrm{E}[X])^2\right] \\ =& \mathrm{E}[X^2] - \mathrm{E}[X]^2\end{aligned}$$

- $\mathrm{Var}(a + bX) = b^2\mathrm{Var}(X)$
- Standard deviation is $\sqrt{\mathrm{Var}(X)}$

Covariance

- Covariance of X and Y:

$$\begin{aligned}\mathrm{Cov}(X,Y) =& \mathrm{E}\left[(X-\mathrm{E}[X])\,(Y-\mathrm{E}[Y])\right]\\ =& \mathrm{E}[XY]-\mathrm{E}[X]\mathrm{E}[Y]\\ =& \mathrm{E}[(X-\mathrm{E}[X])\,Y]=\mathrm{E}[X\,(Y-\mathrm{E}[Y])]\end{aligned}$$

- $\mathrm{Cov}(X,X)=\mathrm{Var}(X)$
- $\mathrm{Cov}(a_1+b_1X+c_1Y, a_2+b_2X+c_2Y) =$
  $b_1b_2\mathrm{Var}(X)+c_1c_2\mathrm{Var}(Y)+(b_1c_2+c_1b_2)\mathrm{Cov}(X,Y)$
- $\mathrm{Var}(a+bX+cY)=b^2\mathrm{Var}(X)+c^2\mathrm{Var}(Y)+2bc\mathrm{Cov}(X,Y)$
- $\mathrm{Var}\left(\sum_{i=1}^{N} b_iX_i\right)=\sum_{i=1}^{N}\left(\sum_{j=1}^{N} b_ib_j\mathrm{Cov}(X_i,X_j)\right)$

Variance

We will often need to calculate the variance of sums like in the last two bullet points. You should make sure you're comfortable with them. It's not too much algebra to derive them from the definition of variance. Let's do so now.

$$\begin{aligned}
\mathrm{Var}(a+bX+cY) =& \mathrm{E}\left[(a+bX+cY-\mathrm{E}[a+bX+cY])^2\right] \\
=& \mathrm{E}\left[(a+bX+cY-(a+b\mathrm{E}[X]+c\mathrm{E}[Y]))^2\right] \\
=& \mathrm{E}\left[(b(X-\mathrm{E}[X])+c(Y-\mathrm{E}[Y]))^2\right] \\
=& \mathrm{E}\left[b^2(X-\mathrm{E}[X])^2+c^2(Y-\mathrm{E}[Y])^2+2bc(X-\mathrm{E}[X])(Y-\mathrm{E}[Y])\right] \\
=& b^2\mathrm{E}[(X-\mathrm{E}[X])^2]+c^2\mathrm{E}[(Y-\mathrm{E}[Y])^2]+2bc\mathrm{E}\left[(X-\mathrm{E}[X])(Y-\mathrm{E}[Y])\right] \\
=& b^2\mathrm{Var}(X)+c^2\mathrm{Var}(Y)+2bc\mathrm{Cov}(X,Y)
\end{aligned}$$

The last bullet with the summation follows from similar logic.

Correlation

- Correlation of X and Y is $\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}$
- Cauchy-Schwartz inequality: $|\mathrm{Cov}(X, Y)| \leq \sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}$
  - Implies $-1 \leq \mathrm{Corr}(X, Y) \leq 1$
- $\mathrm{Corr}(X, Y) = \pm 1$ iff $Y = a + bX$

Conditional expectation

back to health insurance example

$E[Y_{i1}|D_i=1]$ $E[Y_{i0}|D_i=0]$

$E[Y_{i0}|D_i=1]$ → counterfactual

- Conditional expectation of Y given $X = x$:

$$\mathrm{E}[Y|X=x] = \begin{cases} \sum_{i=1}^{m} y_i f_{Y|X}(y_i|x) & \text{if discrete} \\ \int_{-\infty}^{\infty} y f_{Y|X}(y|x)dy & \text{if continuous} \end{cases}$$

- $\mathrm{E}[Y|X]$ is a function of X
- Has all properties of unconditional expectation
- Properties:
  - $\mathrm{E}[g_1(X) + g_2(X)Y|X] = g_1(X) + g_2(X)\mathrm{E}[Y|X]$ (把X当作数)
  - Law of iterated expectations: $\mathrm{E}_X\left[\mathrm{E}_{Y|X}[Y|X]\right] = \mathrm{E}[Y]$ (不考证明)
  - $\mathrm{E}\left[X\left(Y - \mathrm{E}[Y|X]\right)\right] = 0$

$E_{(Y|X)}$：X为自变量

$\mathrm{E}[g(x)(Y-\mathrm{E}(Y|X))]=0$
$=\mathrm{E}[\mathrm{E}[g(x)(Y-\mathrm{E}(Y|X)|X]]$
$=\mathrm{E}[g(x)\mathrm{E}[Y-\mathrm{E}(Y|X)]]$
$=0$

Conditional variance

- Conditional variance: $\mathrm{Var}(Y|X) = \mathrm{E}\left[(Y - \mathrm{E}[Y|X])^2 \,|X\right]$
- Relation to variance:

$$\mathrm{Var}(Y) = \mathrm{E}_X\left[\mathrm{Var}(Y|X)\right] + \mathrm{Var}\left(\mathrm{E}[Y|X]\right)$$

  - Analysis of variance (ANOVA)
  - $\mathrm{E}_X\left[\mathrm{Var}(Y|X)\right]$ is within-X variance
  - $\mathrm{Var}\left(\mathrm{E}[Y|X]\right)$ is between-X variance

Random Variables

Toss a coin: Random events Values Random Variable (R. V.)

$$\left.\begin{array}{ll} H & 0 \\ T & 1 \end{array}\right\} \Rightarrow X$$

Definiton

Let S be the sample space for an experiment. A real-valued function that assigns one and only one real number $X(s) = x$ to each element $s \in S$ is called a **random variable**.

When a probability measure has been specified on the sample space of an experiment, we can determine the probability associated with $X(s)$.

Distribution

Definition

Let $C \subset \mathbb{R}$. Let X be a random variable. The **distribution** *of X is the collection of all probabilities of the form $P(X \in C) \equiv P(\{s : X(s) \in C\})$ for all sets C s.t. $\{x \in C\}$ is an event.*

Example: Toss a fair coin 10 times. $X \equiv$ number of heads obtained.

Possible values of X: 0, 1, 2, ..., 10

$\Rightarrow P(X = x) =$ sum of the prob of all outcomes in the event $\{X = x\}$, i.e. get x heads.

Discrete Distributions

Definition

A random variable has a **discrete distribution** *or X is a* **discrete random variable** *if X can take only a finite number k of different values $x_1, \ldots, x_k$ or, at most, an infinite sequence of different values $x_1, x_2, \ldots$ (countably infinite)*

Definition

If a random variable X has a discrete distribution, the **probability function (p.f.)** *of X is defined as the function f s.t. for $\forall x \in \mathbb{R}$*

$$f(x) = P(X = x).$$

We need: $0 \leq f(x) \leq 1$, $\sum f(x) = 1$.

Some textbook call it **probability mass function (pmf)**.

Continuous Distributions

- R.V. that can assume every value in an interval
- Only talk about probability of X for a given interval

Definition

A r.v. X has a continuous distribution or X is a **continuous r.v.** *if $\exists$ nonnegative function f, defined on $\mathbb{R}$, s.t. for every interval of real numbers, the probability that X takes a value in the interval is the integral:*

$$P(a \leq X \leq b) = \int_a^b f(x)dx.$$

Similarly, $P(X \geq a) = \int_a^\infty f(x)dx$; $P(X \leq b) = \int_{-\infty}^b f(x)dx$.

Definition

If X has a continuous distribution, the function f in above definition is called the **probability density function**. *(p.d.f.) of X.*

Remark: for any pdf,

- $f(x) \geq 0, \ \forall x \in \mathbb{R}$
- $\int_{-\infty}^{\infty} f(x)dx = 1$
- $P(X = x) = 0$ for a continuous r.v.

Example:

$$f(x) = \begin{cases} 3x^2, & 0 \leq x \leq 1 \\ 0, & o.w. \end{cases}$$

$P(0 \leq x \leq 0.5) =$

Cumulative Distribution Function

Definition

The **cumulative distribution function (c.d.f)** F *of a r.v* X *(either discrete or continuous) is the function:* $F(x) = P(X \leq x)$ *for* $x \in \mathbb{R}$.

Example: (c.d.f. of Bernoulli Distribution)

$$\begin{array}{l} P(X=0) = 1-p \\ P(X=1) = p \end{array} \Rightarrow F(x) = \begin{cases} 0, & x < 0 \\ 1-p, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$

What else?

- Normal distribution
- Student's t-distribution
- Law of Large Number and Central Limit Theorem
- Testing
- Confidence interval
- ...

No worries! We will cover these when we need them.